Sample Bus Specification (excerpt for demo)

The write data, which is driven by the controller, is latched into the destination register at the rising edge of the clock.

Although the transfer is initiated at the falling edge, the data is not sampled by the slave until the arbiter grants ownership of the bus to the requester.

Each buffer that stores incoming data is flushed when the engine completes an operation, thereby reducing the overall latency of the pipeline.

The address decoding logic, implemented within the arbiter, determines which slave owns the bus when multiple masters present conflicting requests simultaneously.

Because the register file is accessed through a single read port, the controller must stall the pipeline whenever two concurrent reads target the same word.

A speculative fetch improves the throughput of the processor, but the speculative results are discarded if a branch prediction is subsequently found to be incorrect.

Notwithstanding the reserved field, the protocol mandates that every pending transaction be completed before the bridge is allowed to enter the low-power state.

The data, whose upper and lower halves are aggregated into a single continuous burst, is transmitted across the asynchronous boundary without any additional buffering.

Command Interface Signals

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Write Data and Read Data Signals

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Low Power Control and Status

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